

Introduction: What is SPSS?

Lecture no 76

- Originally it is an acronym of Statistical Package for the Social Science but now it stands for Statistical Product and Service Solutions
- One of the most popular statistical packages which can perform highly complex data manipulation and analysis with simple instructions

The Four Windows:

Data editor

Output viewer

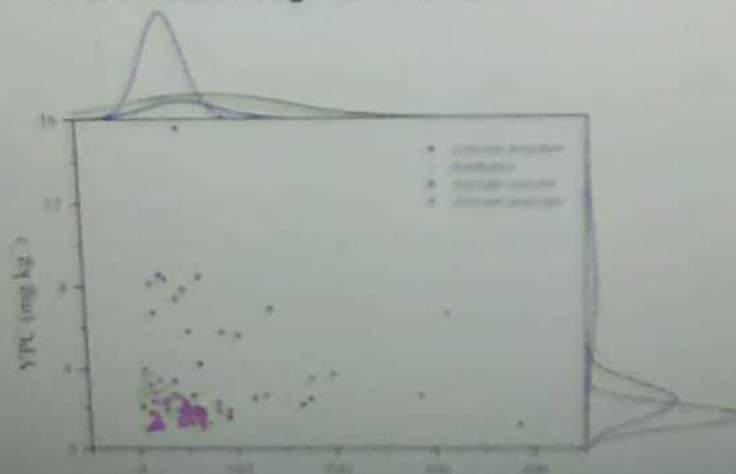
Syntax editor

Script window

Measures of Dispersion

PPO = Polyphenol Oxidase activity

YPC = Yellow Pigment Content.



<https://www.ncbi.nlm.nih.gov/articles/PMC3198538>

Measures of dispersion - NCBI

Search for: What is the most commonly used dispersion?

What are the 4 measures of dispersion?

4 Commonly Used Measures of Dispersion | Statistics

- Measure # 1. Range
- Measure # 2. Quartile Deviation
- Measure # 3. Average Deviation (A.D.) or Mean Deviation (M.D.)
- Measure # 4. Standard Deviation or S.D. and Variance

<https://www.yourarticlelibrary.com/statistics-2-4-com>

4 Commonly Used Measures of Dispersion | Statistics

uses of dispersion in statistics



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The measures of dispersion are used **for defining the data spread or its variation around a central value**. Two different samples may have an equal mean or median, but completely different variability levels, or vice versa. A proper description of a data set should include both of these characteristics. 5 days ago

<https://cleartax.in> > home > Economy > Economics

Definition, Latest News, and Why Dispersion is Important?

About featured snippets Feedback

Types of Probability Sampling Designs

- Simple random sampling
- Stratified sampling
- Systematic sampling
- Cluster (area) sampling
- Multistage sampling

Interquartile Range

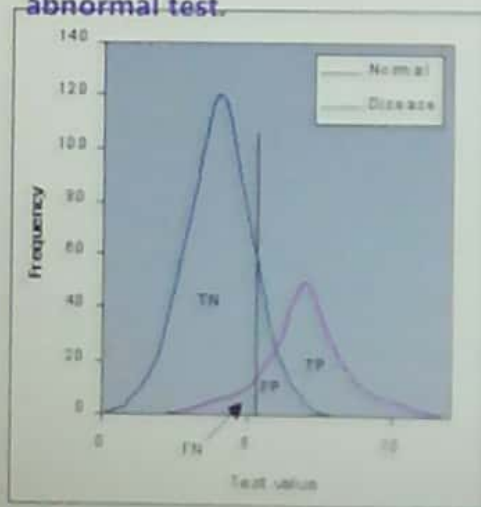
The **interquartile range** is the difference between the upper quartile and the lower quartile.

$$14 - 8.5 = 5.5$$

ROC CURVES

The **Sensitivity** and **Specificity** of a diagnostic test depends on more than just the "quality" of the test.

They also depends on the definition of what constitutes an **abnormal test**.



Look at the **idealized graph** showing the number of **patients with and without a disease** arranged according to the value of a **diagnostic test**.

The **area of overlap** indicates where the test **cannot distinguish** normal from disease.

In practice, we choose a **cutpoint** above which we consider the test to be **abnormal** and below which we consider the test to be **normal**.

Spearman Rank **Correlation Coefficient** **(r_s)**

- It is a non-parametric measure of **correlation**.
- This procedure makes use of the two sets of ranks that may be assigned to the sample values of x and Y.
- Spearman Rank **correlation** coefficient could be computed in the following cases:
 - Both variables are quantitative.
 - Both variables are qualitative ordinal.
 - One variable is quantitative and the other is qualitative ordinal.

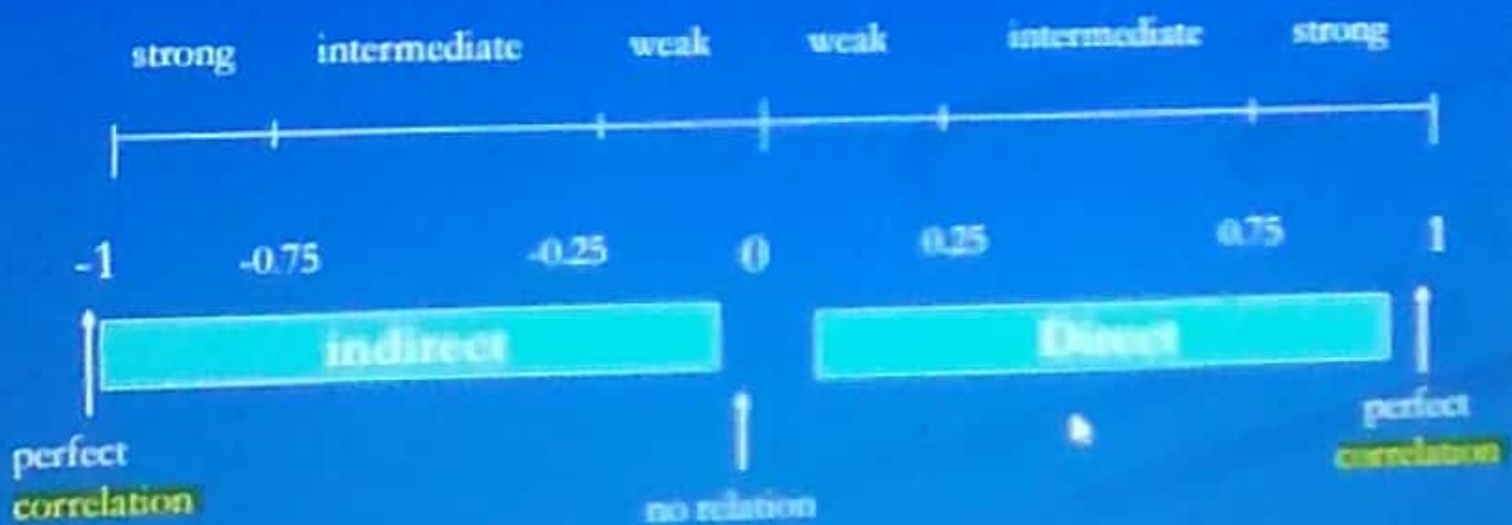
Correlation and Regression

- **Correlation** describes the strength of a linear relationship between two variables
- Linear means "straight line"
- Regression tells us how to draw the straight line described by the **correlation**

Important Terms

- Sporadic:** disease occurs occasionally, irregularly ■
- Endemic:** disease stays in population at low frequency ■
- Epidemic:** sudden outbreak in disease above typical level ■
- Pandemic:** epidemic over wide area (may be entire world). ■
- Morbidity:** all reported cases of disease, illness, and disability ■
- Mortality:** reported deaths due to a disease ■

- The value of r ranges between (-1) and $(+1)$
- The value of r denotes the strength of the association as illustrated by the following diagram.



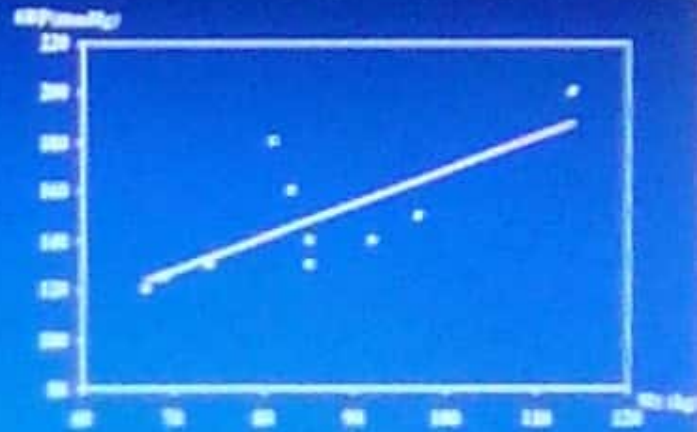
Regression

- Uses a variable (x) to predict some outcome variable (y)
- Tells you how values in y change as a function of changes in values of x

$$\sum x = \frac{(\sum x)}{n}$$

Regression Equation

- Regression equation describes the regression line mathematically
 - Intercept
 - Slope



Multiple Regression

Multiple regression analysis is a straightforward extension of simple regression analysis which allows more than one independent variable.

Measures of Dispersion

Dispersion is defined as the **extent** to which the observations in the dataset are **spread away** from the **central value**.

Measures of Dispersion are the methods that **convey** information regarding the **amount of variability** present in the set of data.

If **all the values** in the data are the **same** then there is **no dispersion**.

Other terms used **synonymously** with **dispersion** includes **variation**, **spread** and **scatter**.

Measures of Dispersion

Fasting Plasma Glucose levels of 7 individuals in two groups are given.

Group 1	Group 2
x	y
100	120
110	105
90	80
105	100
95	95
101	103
99	97

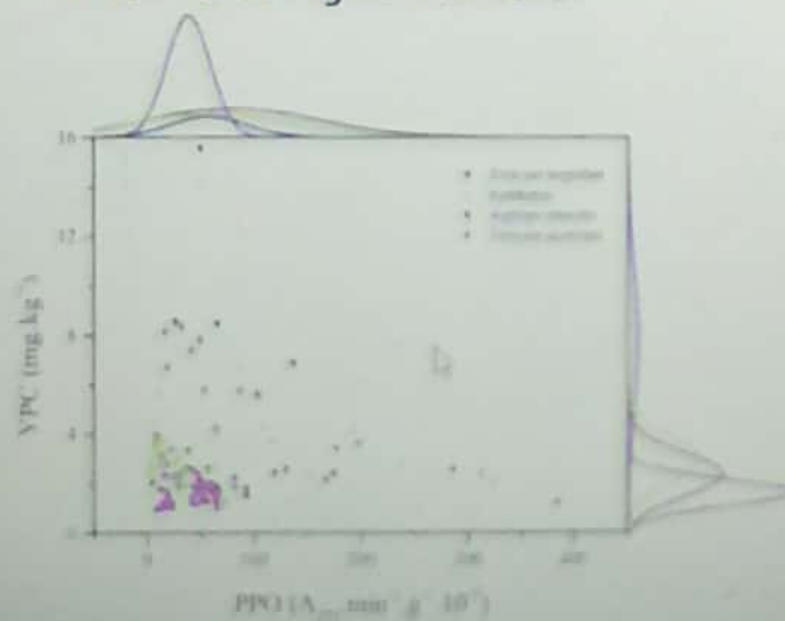
$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{700}{7} = 100$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} = \frac{700}{7} = 100$$

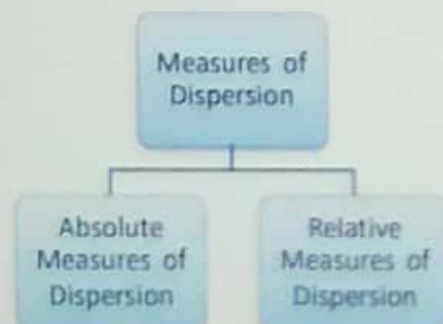
Measures of Dispersion

PPO = Polyphenol Oxidase activity

YPC = Yellow Pigment Content.



Li F. Y et al (2015) "Polyphenol oxidase activity and yellow pigment content in *Aegilops tauschii*, *Triticum Turgidum*, *Triticum aestivum*, synthetic hexaploid wheat and its parents." *Journal of cereal sciences* 107: pp 192 – 201



Measures which are expressed in terms of the same units as the original data are termed as **Absolute Measures of Dispersion**.

Measures which are expressed in terms of the ratios and percentages these measures are independent of the units of measurement and termed as **Relative Measures of Dispersion**.

Absolute Measures of Dispersion are of different types.

- **Range**
- **Quartile Deviation**
- **Mean Absolute Deviation**
- **Standard Deviation and Variance**

Absolute Measures of Dispersion - I

Range

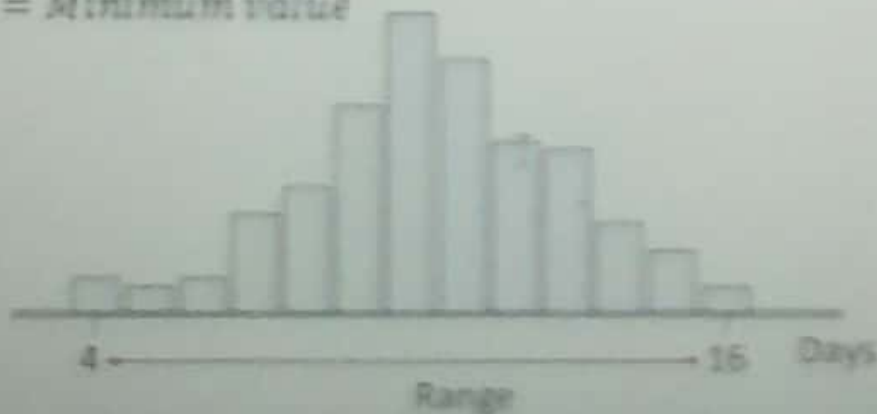
Range is defined as the difference between the maximum and the minimum value in the data.

It is sensitive to only the most extreme values in the list. The range of a list is 0 if and only if all the data-points in the list are equal.

$$\text{Range} = x_m - x_o$$

x_m = Maximum value

x_o = Minimum value



Pros and Cons of Range

- Pros

- best for symmetric data with no outliers
- easy to compute and understand
- good option for ordinal data

- Cons

- doesn't use all of the data, only the extremes
- very much affected if the extremes are outliers
- only shows maximum spread, does not show shape

Absolute Measures of Dispersion are of different types.

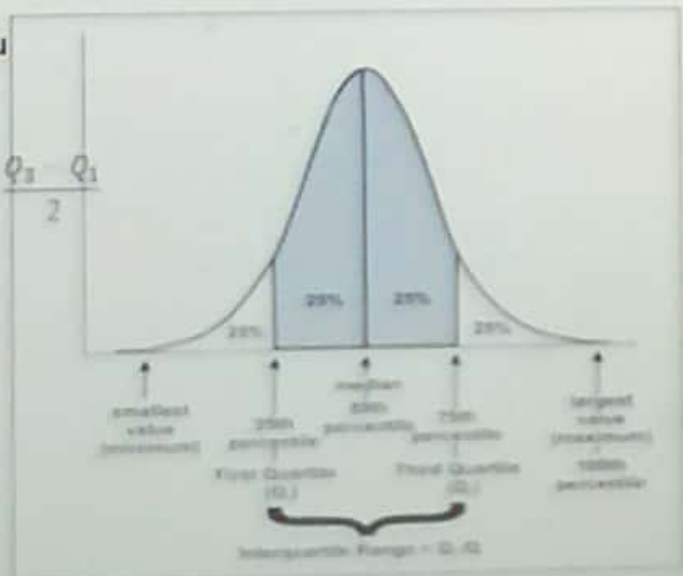
- **Range**
- **Quartile Deviation**
- **Mean Absolute Deviation**
- **Standard Deviation and Variance**

- **Quartile Deviation is the half distance between the Upper Quartile Q_3 (i.e. 75th percentile) and Lower Quartile Q_1 (i.e. 25th Percentile).**
- **Essentially describes that how much the middle 50% of the data varies.**
- **Also known as semi - interqu**

$$Q.D. = \frac{\text{Inter Quartile Range (IQR)}}{2} = \frac{Q_3 - Q_1}{2}$$

$$Q_1 = \left(\frac{1(n+1)}{4} \right)^{\text{th}} \text{ Value}$$

$$Q_3 = \left(\frac{3(n+1)}{4} \right)^{\text{th}} \text{ Value}$$



<https://www.cdr.gov/epidemiology/teaching/1517/lesson/quarter1/lesson10127>

Absolute Measures of Dispersion - I

Method for Determining the Inter - Quartile Range.

Step 1: Arrange the data in and ascending order.

Step 2: Find the position of the lower Quartile i.e. Q_1 and upper Quartile i.e. Q_3 using following formula.

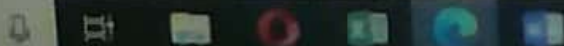
$$Q_1 = \left(\frac{1(n+1)}{4} \right)^{th} \text{ Value}$$

$$Q_3 = \left(\frac{3(n+1)}{4} \right)^{th} \text{ Value}$$

Absolute Measures of Dispersion - I

Properties and uses of the Inter-Quartile Range

- The IQR is generally used in conjunction with the median. Together, they are useful for characterizing the central location and spread of any frequency distribution, but particularly those that are skewed.
- For a more complete characterization of a frequency distribution, the 1st and 3rd quartiles are sometimes used with the minimum value, the median, and the maximum value to produce a five number summary of the distribution.
- Together these five numbers provide a good description of the centre, spread and shape of the distribution.



Desktop

Pros and Cons of Quartile Deviation

- Pros

- Good for ordinal data.
- Good for Skewed data.
- Ignores extreme values
- More stable than the range because it ignores outliers

- Cons

- Harder to calculate and understand
- Doesn't use all the information (ignores half of the data-points, not just the outliers)
 - Tails almost always matter in data and these aren't included
 - Outliers can also sometimes matter and again these aren't included.

Absolute Measures of Dispersion are of different types.

- **Range**
- **Quartile Deviation**
- **Mean Absolute Deviation**
- **Standard Deviation and Variance**

Pros and Cons of Mean Absolute Deviation

• Pros

- Easy to calculate and easy to understand.
- Based on all the observation in the data.
- Shows the scatter of observations from its central value.
- It facilitated comparison between different items in a series.

• Cons

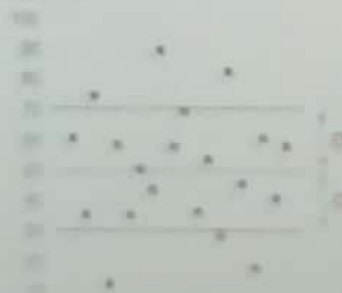
- It violates the algebraic principle by ignoring the signs of the values.
- Affected by fluctuations in the sampling.
- Depending on the type of central value being used it produces different results.

Variance

It is defined as the mean of the squared deviations of the observations from their mean.

If the variance is calculated for the population data then it is denoted by σ^2

$$\sigma^2 = \frac{\sum_{i=1}^N (X_i - \mu)^2}{N}$$



If the variance is calculated for the sample data obtained from a population then it is denoted by:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

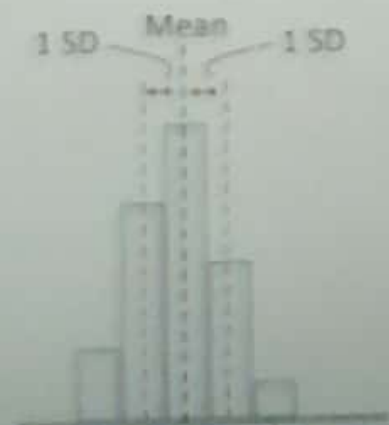
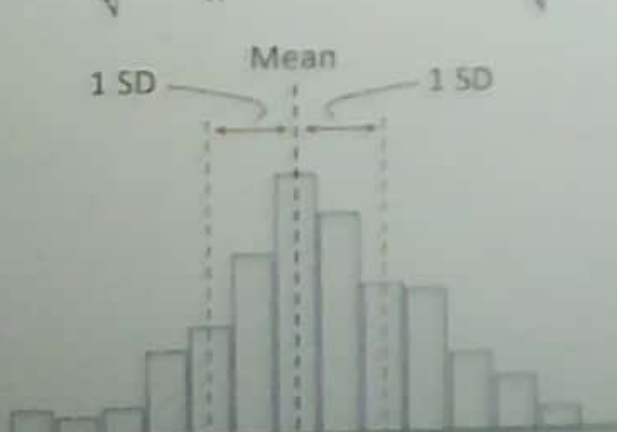
Standard Deviation

- The standard deviation (SD) is the square root of the variance.
 - small SD = values cluster closely around the mean
 - large SD = values are scattered

$$\sigma^2 = \sqrt{\frac{\sum_{i=1}^N (X_i - \mu)^2}{N}}$$

$$S^2 = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

$$s^2 = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$



Example

The incubation period of Hepatitis A is given:

27, 32, 25, 30, 22

Calculate amount of Spread?

Solution:

Step 1: Calculate the Arithmetic Mean $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$

$$\bar{x} = \frac{27 + 31 + 15 + 30 + 22}{5} = \frac{125}{5} = 25.0 \text{ days}$$

Absolute Measures of Dispersion - III

Properties of Variance

1. The variance of a constant is equal to zero.
2. The variance is independent of Origin, i.e. it remains unchanged when a constant is added to or subtracted from the each and every observation of the data.
3. The variance is multiplied or divided by the square of the constant if each observation is multiplied or divided by the same constant.
4. The variance of the sum of difference of two independent variable is equal to the sum of their respective variances.

Relative Measures of Dispersion

Coefficient of Variation

- The Standard Deviation is useful as a measure of variation within a given set of data.
- However, comparing dispersion in two datasets on the basis of standard deviation may lead to fallacious results.
- Two means may be quite different.
- when one desires to compare the dispersion between two or more than two data sets then we prefer using coefficient of variation.

Coefficient of Variation (C.V)

$$C.V. (x) = \frac{s}{\bar{x}} \times 100$$

Suppose two samples of human males yield the following results:

	Sample 1	Sample 2
Age	25 years	11 years
Mean weight	145 pounds	80 pounds
Standard deviation	40 pounds	10 pounds

We wish to know which is more variable, the weights of the 25-year-olds or the weights of the 11-year-olds.

Relative Measures of Dispersion

Properties of Coefficient of Variation

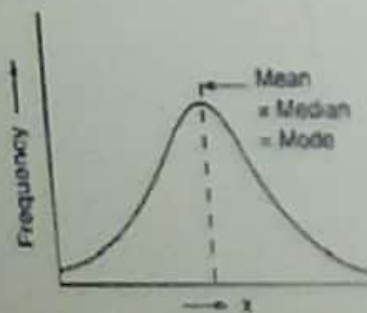
- The Coefficient of Variation is a useful measure, when comparing the results obtained by different persons, who are conducting investigations involving same variable.
- Since the coefficient of Variation is independent of the scale of measurement, it is useful statistic for comparing the variability of two or more variables measured on different scales.

for example, using the coefficient of variation to compare the variability in weights of one sample of subjects whose weights are expressed in pounds with the variability in weights of another sample of subjects whose weights are expressed in kilograms.

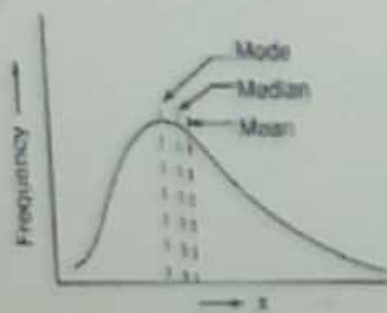
Relative Measures of Dispersion

Skewness

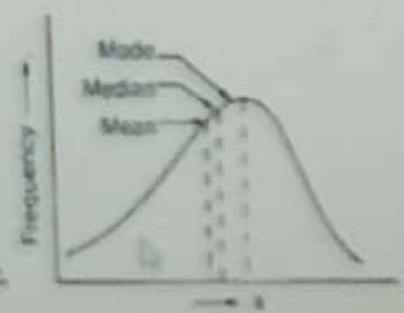
"It is defined as the departure from Symmetry"



(a) Symmetrical distribution



(b) Positively skewed distribution



(c) Negatively skewed distribution

Measures of Skewness

Coefficient of Skewness help us to measure amount of the departure from symmetry.

It is denoted by Sk

Karl Pearson (1857 – 1936)

$$Sk = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}}$$

Measures of Skewness

Since we know that mode is sometimes ill defined to located by simple methods.

It is replaced by its equivalent for moderately skewed distributions

$$Sk = \frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

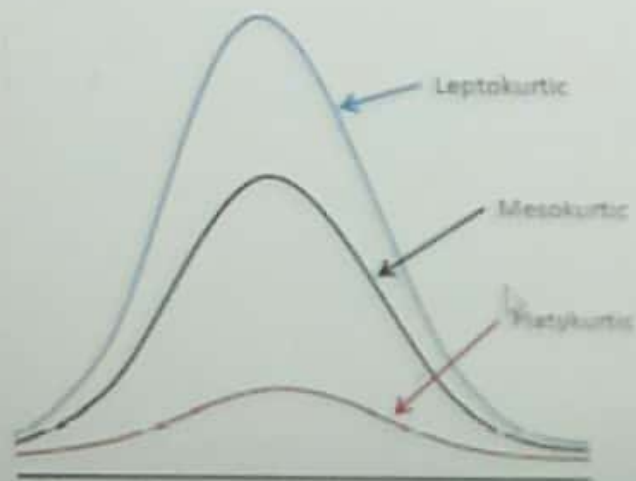
Arthur Lyon Bowley (1869 – 1957)

$$Sk = \frac{Q_1 + Q_3 - 2Q_2}{Q_3 - Q_1}$$

Relative Measures of Dispersion

Kurtosis

Peakedness of the data



<https://www.logosheikh.org/wiki/File/Kurtosis.jpg>

Measures of Skewness

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Skewness and Kurtosis

Measures of Skewness

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Arthur Lyon Bowley (1869 – 1957)

$$Sk = \frac{Q_1 + Q_3 - 2Q_2}{Q_3 - Q_1}$$

Measures of Skewness

Qualitative Variables:

**There is not need to
measure coefficient of
skewness**

Quantitative Variables

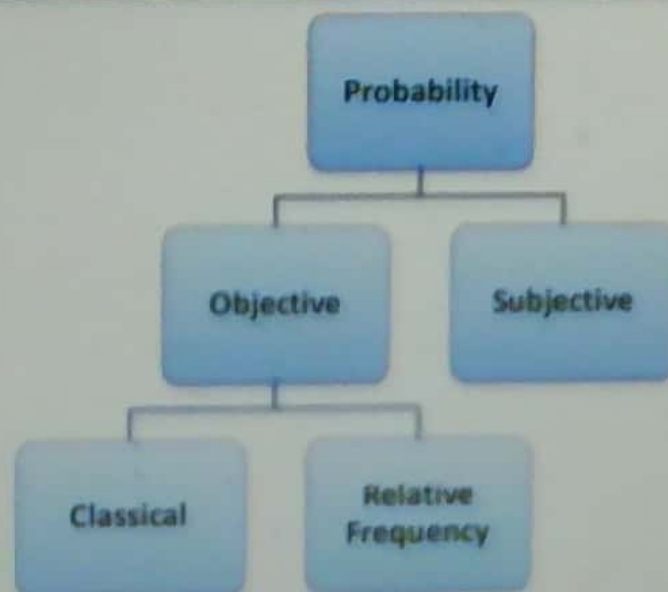
- **Frequency Histogram**
- **Frequency Curve**
- **Comparison of Mean,
Median and Mode,**
- **Coefficients of Skewness**

There are various other types of events like

- **Mutually Exclusive Events**
- **Collectively Exhaustive Events**
- **Equally Likely Events**
- **Independent and Dependent Events**

END

Introduction to Probability - II



In the early 1950s, L.J. Savage gave considerable impetus to what is called the "Personalistic" or subjective concept of Probability.

The Subjective definition of probability utilizes intuition, experience, and collective wisdom to assign a degree of belief that an event will occur.

Classical Probability (*Priori*)

- This concept dates back to 17th century and the work of two mathematicians Pascal and Fermat.
- Much of this theory was developed out of the attempts to solve problems related to the games of chance, such as those involving the rolling of dice.

DEFINITION

If an event can occur in N mutually exclusive and equally likely ways, and if m of these possess a trait E , the probability of the occurrence of E is equal to m/N .

If we read $P(E)$ as "the probability of E ," we may express this definition as

$$P(E) = \frac{m}{N}$$

Elementary Properties of Probability

In 1933 the axiomatic approach to probability was formalized by the Russian mathematician A.N. Kolmogorov.

The basis of this approach is embodied in three properties from which a whole system of probability theory is constructed through the use of mathematical logic.



https://en.wikipedia.org/wiki/Andrey_Kolmogorov

Axioms of Probability

Elementary Properties of Probability

There are three axiomatic properties of Probability

- Axiom of Non - Negativity
- Axiom of Exhaustiveness
- Axiom of Additive - ness

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Diagnostic Testing - I

Some Common Screening Tests

- Fasting blood cholesterol for heart disease.
- Fasting blood sugar for diabetes.
- Blood Pressure for hypertension.
- Mammography for breast cancer.
- PSA test for prostate cancer.
- Fecal Occult blood test for colon cancer.

Downloaded from Studydrive

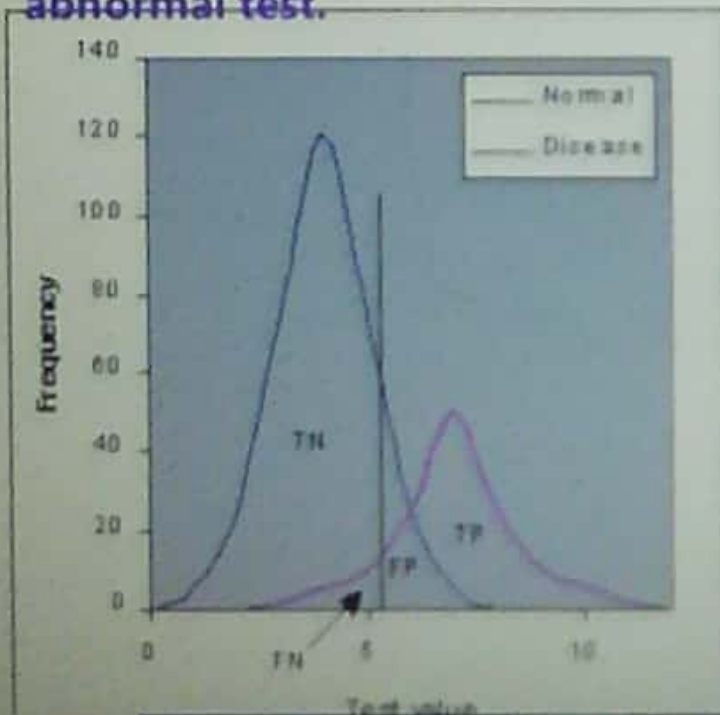
Variation in Biological Values

- Many test results have a continuous scale (are continuous variables)
 - Blood Glucose level (70 – 100 mg/dL)
 - Cholesterol level (less than 100 mg/dL)
 - Blood Pressure
 - Systolic Blood Pressure 120
 - Diastolic Blood Pressure 80
- Distribution of Biological measurements in humans may or may not permit easy separation of diseased from non-diseased individuals, based upon the value of the measurements.

ROC CURVES

The **Sensitivity** and **Specificity** of a diagnostic test depends on more than just the "quality" of the test.

They also depends on the definition of what constitutes an **abnormal test**.



Look at the **idealized graph** showing the number of **patients with and without a disease** arranged according to the value of a **diagnostic test**.

The **area of overlap** indicates where the **test cannot distinguish normal from disease**.

In practice, we choose a **cutpoint** above which we consider the test to be **abnormal** and below which we consider the test to be **normal**.

Properties and Uses of Incidence Rate

An incidence rate describes how quickly disease occurs in a population

Because person-time is calculated for each subject, it can accommodate persons coming into and leaving the study.

The denominator accounts for study participants who are lost to follow-up or who die during the study period.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4888888/>

Measures of Morbidity - I